Probability Distributions KEY

Name _____

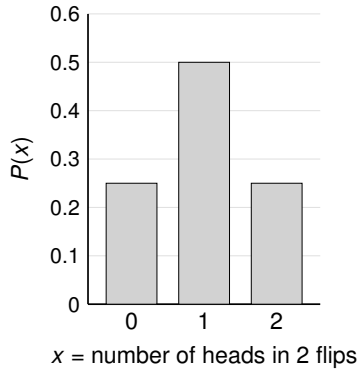
Flip a coin 2 times. Let X = the number of heads. X is a **random variable**: a number that comes from a chance process.

Outcomes: HH, HT, TH, TT (all equally likely)

x (heads)	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$

This table is the **probability distribution** of X. It lists every value of X and its probability.

$P(x)$ means the probability that X equals x. Capital X is the random variable. Small x is one of its values.



A valid probability distribution:
1. Every probability is from 0 to 1. 2. The probabilities add to 1: $\Sigma P(x) = 1$

Check: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. Valid.

Using a distribution. Add the probabilities of the values you want.

$P(X \geq 1) = P(1) + P(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ ($\geq$ means "greater than or equal to")

Theoretical or empirical?

Theoretical	From reasoning about the chance process (equally likely outcomes, or the multiplication rule), like the coin flips above.
Empirical	From data. Divide each count by the total. 50 families: 0 cars: 5, 1 car: 20, 2 cars: 18, 3 cars: 7 $P(0) = \frac{5}{50} = 0.10$, $P(1) = 0.40$, $P(2) = 0.36$, $P(3) = 0.14$

Work with a partner. Compare answers when you finish each problem.

1. Is each table a valid probability distribution? Explain.

(a)

x	1	2	3
$P(x)$	0.2	0.3	0.5

(b)

x	1	2	3
$P(x)$	0.4	0.4	0.3

(c)

x	1	2	3
$P(x)$	0.5	-0.1	0.6

(d)

x	1	2	3
$P(x)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

(a) Valid: sums to 1. (b) Not valid: sums to 1.1. (c) Not valid: a probability cannot be negative.
(d) Valid: each is from 0 to 1, and they sum to 1.

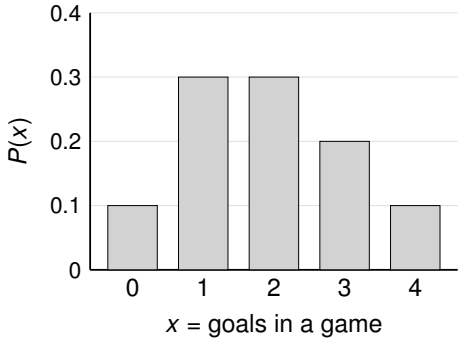
2. Find the missing probability.

x	0	1	2	3
$P(x)$	0.15	0.35	?	0.10

$1 - (0.15 + 0.35 + 0.10) = 0.40$

3. This is the distribution of the number of goals a soccer team scores in a game.

x	0	1	2	3	4
$P(x)$	0.1	0.3	0.3	0.2	0.1



(a) $P(X \geq 2)$ $0.3 + 0.2 + 0.1 = 0.6$

(b) $P(X < 1)$ 0.1

(c) $P(1 \leq X \leq 3)$ $0.3 + 0.3 + 0.2 = 0.8$

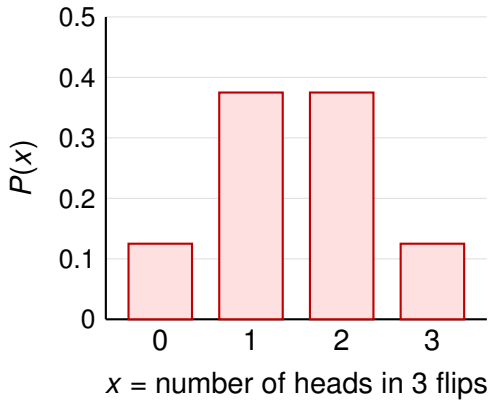
4. Flip a coin 3 times. Let X = the number of heads.

   (a) List all 8 outcomes. HHH, HHT, HTH, THH, HTT, THT, TTH, TTT

   (b) Make the probability distribution of X.

x	0	1	2	3
$P(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

   (c) Draw a bar graph of the distribution.



   (d) Find $P(X \geq 2)$. $\frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$

5. A player makes 70% of her free throws. She shoots 2. Her shots do not affect each other. Let X = the number she makes. Make the probability distribution of X.

x	0	1	2
$P(x)$	$0.3 \cdot 0.3 = 0.09$	$2(0.7)(0.3) = 0.42$	$0.7 \cdot 0.7 = 0.49$

   $P(1)$ has a 2: she can make then miss, or miss then make. Check: $0.09 + 0.42 + 0.49 = 1$.

6. 40 households said how many pets they have: 0 pets: 12, 1 pet: 16, 2 pets: 8, 3 pets: 4.

   (a) Make the empirical probability distribution.
   $P(0) = 0.30$, $P(1) = 0.40$, $P(2) = 0.20$, $P(3) = 0.10$

   (b) Find P(at least 1 pet). $0.40 + 0.20 + 0.10 = 0.70$ (or $1 - 0.30$)